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Tribological synergies among chemical-modified graphene oxide nanomaterials and a phosphonium ionic liquid as additives of a biolubricant

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ABSTRACT

In the present work, antifriction and antiwear synergies of two functionalized graphene oxides (reduced graphene oxide, rGO, and reduced graphene oxide modified with octadecylamine, rGO@ODA) with a phosphonium ionic liquid (IL) as additives of a biodegradable ester base oil, BIOE, were investigated. For this aim, four BIOE nanodispersions were formulated: two nanodispersions without IL and two others with IL (hybrid nanolubricants), being the nanodispersions: BIOE + 0.05 wt% rGO, BIOE + 0.05 wt% rGO@ODA, BIOE + 1 wt% IL + 0.05 wt% rGO and BIOE + 1 wt% IL + 0.05 wt% rGO@ODA, and all of them showing good temporal stability (at least 3 weeks), especially those containing the ionic liquid. Tribological tests, under pure sliding conditions, were made with the aforementioned nanolubricants and with neat BIOE under a 20 N working load. All hybrid and non-hybrid nanolubricants resulted in lower friction coefficients compared to BIOE oil, with the largest friction reduction of 34% achieved by the 1 wt% IL + 0.05 wt% rGO nanolubricant. Discs lubricated with the prepared nanolubricants showed less wear than discs lubricated with BIOE, achieving the highest reduction (34%) in wear track width also for the 1 wt% IL + 0.05 wt% rGO nanolubricant. In addition, through confocal Raman microscopy, as well as roughness analysis of tested discs, it can be concluded that surface repairing, synergistic effect and tribofilm formation mechanisms occur.

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1. Introduction

One of the most important problems in science, technology and economy is planning long-term resource consumption and social trends as well as finding an appropriate balance between them. Friction consumes around 20% of all energy used worldwide [1]. Several studies show that the use of suitable lubricants could reduce global energy consumption, greenhouse gas emissions, as well as extend the useful life of machinery [1,2]. The International Energy Agency (IEA) estimated that the main contribution (38%) of key technologies to reduce CO_2 emissions will come from end-use energy efficiency [2]. In addition, the loss of lubricants is a major environmental problem: soil and water are immediately altered by failures in the lubrication system while the air is contaminated with volatile lubricants [3]. Therefore, it is essential to develop optimized lubricants to minimize energy losses which are also eco-friendly, reducing its environmental impact. Generally, a lubricant is composed by approximately 90% of base oil and 10% of additives. Thus, to find the aforementioned energy and environment solutions, it is important to analyze both lubricant components.

In this vein, a possible way to design an optimal lubricant is to use biodegradable oils instead of traditional mineral oils that are harmful to the environment. In fact, several authors have also emphasized the need for biodegradable lubricants to replace traditional lubricants [4,5]. However, some properties of biodegradable lubricants are slightly worse than those of mineral oils. The use of additives can lead to significant enhancements in the tribological properties of lubricants [3]. Therefore, the use of biodegradable oils in combination with appropriate additives can lead to good lubricant properties together with low environmental impact [6]. In last years, nanomaterials and ionic liquids (ILs) have emerged as interesting alternatives to replace conventional additives that

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are harmful to the environment due to adverse emissions (zinc dialkyldithiophosphate, ZDDP) with less toxic additives that also have better tribological properties due to their extraordinary physical and chemical characteristics [7,8]. Specifically, nanoparticles, due to their chemical stability compared to conventional additives, are more durable [9]. Regarding ILs, Oulego et al. [10] compared environmental properties of 12 ILs potentially used as lubricant additives (containing phosphonium-, imidazolium- and ammonium-based cations) with two different ZDDPs. These authors observed that the tributylethylphosphonium diethylphosphate IL was the least toxic, with EC_{50} values of 55.2 ± 0.7 mg/L (toxic units, TU = 1.81) and 39.1 ± 0.6 mg/L (TU = 2.56), for *Vibrio fischeri* and *E. coli*, respectively, whereas the EC_{50} values of ZDDP1 and ZDDP2 were usually less than 1.0 ± 0.2 mg/L (TU from 100 to 125) for both bacteria, showing the high toxicity of these traditional additives.

Recently, several researches [11–17] showed that adding small amounts of nanoparticles to lubricants can substantially decrease both friction and wear owing to nanoparticles can enter in the tribological contact and improve the tribofilm performance [18]. Nonetheless, the key challenge in development of nanolubricants is to improve their stability, since agglomerates and sediments are often formed because of van der Waals forces [19]. Hence, the compatibility of the base oil and nanoparticles is a vital issue that must be considered. For this reason, in recent years two different strategies have been addressed. The first one is to incorporate a dispersant with the nanoparticles, in order to ensure their dispersion in the lubricant, and the second one is the chemical modification of the nanoparticle surface to achieve good compatibility with the oil chemical structure.

ILs are one of the most promising dispersants for nanoparticles since, in addition to their interesting dispersing properties [20], they also act as antifriction and anti-wear additives [21,22], exhibiting interesting positive tribological synergies with nanoparticles [23–26]. Nevertheless, research with these hybrid additives (ILs and nanoparticles) for lubricants remains scarce. One of the first examples of graphene nanoparticles in combination with an IL as lubricant additives was published in 2017 by Sanes et al. [25], who examined the tribological behavior of two oils, an isoparaffinic and a fully formulated oil, additived with 1-octyl-3-methylimidazolium tetrafluoroborate and graphene, finding positive anti-wear synergies for the first oil. Moreover, Senatore et al. [26] analyzed the tribological synergies of graphene oxide (GO) and 1-ethyl-3-methylimidazolium IL as additives of a polyalkylene glycol oil, achieving a friction coefficient reduction of 20%.

Regarding the second strategy, surface modification of nanoparticles, there are different types attending to the method used to functionalize the nanoparticle surface [27]: reduction or oxidation [28,29], amide linkage [30,31], silanization [32,33], radical polymerization by surface-induced atom-transfer [34,35] or nanoscale ionic material, among others [36]. For instance, in a previous work [28] the tribological performance of reduced graphene oxide as an additive to an ester base oil was analyzed, finding good tribological results as well as good stability against sedimentation. Moreover, Mungse et al. [31] examined the tribological performance of a 10 W-40 commercial oil additived with alkylated graphene oxide. These authors found stabilities up to 1 month as well as a significantly reduction in both friction and wear compared to 10 W-40 lubricant. Recently, Chen et al. [27] reviewed the temporal stability of numerous nanolubricants, analyzing the different strategies to improve this property, concluding that the surface modification method is key to efficiently disperse the nanoparticles within the lubricant.

Given the above problems, it is clear that nanoparticles as lubricant additives have promising tribological properties, but there is still the problem of the stability of nanolubricants. In this work, the key objective is to examine the tribological properties of two chemically modified nanomaterials, reduced graphene oxide (rGO) and reduced graphene oxide functionalized with octadecylamine (rGO@ODA), separately combined with a phosphonium ionic liquid (tri(butyl)ethylphosphonium diethylphosphate) as hybrid additives of a biodegradable ester base oil (BIOE) at pure sliding conditions. To the best of our knowledge, this is the first research that combines functionalized nanomaterials with the use of ionic liquids as dispersants to improve both the nanolubricants stability and the tribological properties.

2. Materials and methods

2.1. Materials

BIOE is a synthetic biodegradable and polymeric ester supplied by Verkol Lubricantes (Spain). BIOE oil has a density of 0.9318 $g{\cdot}cm^{-3}$ (313.15 K), dynamic viscosity of 485.8 mPa s (313.15 K) and a 155.4 viscosity index. The BIOE oil was characterized through infrared spectroscopy (FTIR, Fig. 1) observing the following peaks: the carbonyl stretching at 1743 cm^{-1}, a peak at 1149 cm^{-1} corresponding to the characteristic absorption of hydroxyl group and carbon–hydrogen (C-H) bending and stretching that are detected at 1459 cm^{-1} and 2921–2854 cm^{-1}, respectively [37].

Tri(butyl)ethylphosphonium diethylphosphate ($[P_{4,4,4,2}][C_2C_2PO_4]$, CAS Number: 20445–94–7) with a purity > 95% was supplied by Iolitec (GmbH, Heilbronn, Germany) and has a kinematic viscosity (40 °C) of 225 cSt as well as a 82 viscosity index [38]. This IL was earlier characterized by FTIR and Raman spectroscopy observing the typical band centered at 1148 cm^{-1} of the P = O stretching mode [24].

Reduced graphene oxide (rGO) nanopowders were synthesized following the Li et al. [29] procedure. For this synthesis, monolayer graphene oxide (GO), provided by Nanoninnova Technologies (Spain), was used as starting material. GO was mixed with KOH and ethanol, and then, this mix was crushed by a mortar and heated at 700 °C in a heating tube furnace. Then, the resulting nanopowder has been washed with deionized water, dried (80 °C) and after all milled in a mortar in order to decrease the size of particles. Detailed information on the experimental procedure to yield rGO can be found in a previous work [28].

Octadecylamine-modified reduced graphene-oxide (rGO@ODA) nanopowders were synthesized following the method reported by Mungse et al. [31] but using graphene oxide (GO), provided by Nanoninnova Technologies (Spain), as starting substance instead of graphite. GO nanopowders (5 g) reduction was carried out by addition of hydrazine monohydrate, (10 mL) and heating the mixture under reflux for 24 h. After letting the mixture reach room temperature, the mixture was divided into two different phases: the lower one contains the precipitated rGO and the upper one is an aqueous phase. Separation of the phase containing the rGO using a separatory funnel followed by filtration through a membrane filter (0.1 μm PTFE membrane) allowed the isolation of rGO. The resulting rGO was washed by means of deionized water (45 mL) to eliminate the residual hydrazine. Afterwards, the acquired rGO powder was mildly oxidized using nitric acid, (6.25 mL, 65% purity) as oxidant, introducing carboxylic groups (–COOH) at the defect sites and edges of rGO [39]. The obtained mild oxidized rGO was refluxed in 30 mL of thionyl chloride, in the presence of 15 mL of N, N-dimethyl-formamide, converting the carboxylic groups into acyl chlorides. The residual thionyl chloride and the acylated-rGO were removed by distillation and washed with deionized water. Later, acylated-rGO powder (3.5 g) was reacted with octadecylamine (ODA, 4.5 g), under argon

atmosphere at 120 °C for 4 days, achieving ODA – rGO powders through an amide linkage. The synthesized powder was dispersed and washed with ethanol (3 × 5 mL) and filtered through membrane obtaining 0.75 g of rGO@ODA.

2.2. Characterization of nanomaterials

GO, rGO and rGO@ODA nanopowders have been characterized through scanning electron microscopy (SEM) to analyze their morphology and examine the structural changes happened after GO chemical modification to produce rGO and rGO@ODA. SEM images (Fig. 2) were obtained with a Zeiss Ultraplus Field Emission Scanning Electron Microscope, FESEM. These microimages reveal that rGO sheets are partly exfoliated (Fig. 2b), in comparison to GO nanosheets (Fig. 2a). On the other hand, rGO@ODA sample (Fig. 2c) presents high aggregation level with crumpled features, this fact is owing to the absence of oxygen groups in the plane of the carbon structure sp^2 allowing ODA – rGO sheets to be connected through van der Waals forces [31].

Furthermore, to confirm the successful nanopowders chemical synthesis, infrared spectroscopy (FTIR) was also used. Fig. 3 shows that GO exhibits strong absorptions bands around 3400 cm^{-1} (stretching O – H of hydroxyl groups), 1730 cm^{-1} (C = O stretch assigned to carboxyl and carbonyl groups), 1620 cm^{-1} (C = C stretch of non-oxidized sp^2 carbon domains), 1370 cm^{-1} (bending O – H), 1250 cm^{-1} (C – O stretch associated to ether groups and phenols) and 1060 cm^{-1} (C – O stretching of hydroxyl groups) [40,41]. These vibration bands show the existence of functional groups of oxygen like carbonyl, carboxyl, or hydroxyl, among others.

Regarding the rGO FTIR spectrum (Fig. 3) the bands of oxygen functional groups appear with lower intensity than in the GO spectrum, being the most important the following ones: around 1730 cm^{-1} C = O (carbonyl/carboxy) and at 1060 cm^{-1} C-O (alkoxy groups). This spectrum variation in comparison to GO spectrum implies a clear reduction of oxygen content in the rGO nanopowders.

Concerning the FTIR spectrum of rGO@ODA nanopowders [42], the addition of octadecylamine through the formation of an amide linkage is clearly verified through the presence of strong vibrational bands centered at 2915 and 2860 cm^{-1}, ascribed to the methylene (C–H) symmetric and asymmetric stretches. Likewise, as in the previous case, the loss of vibration modes related to oxygen containing groups proves the loss of oxygen functional groups as well as the restoration of the graphitic structure [30]. This fact is further supported by strong vibrational bands around 1700 cm^{-1} (amide group C = O stretching), 1580 cm^{-1} (associated to an overlapped N – H bond and sp^2 carbon domain) and finally around 1220 cm^{-1} (stretching vibrational mode of C – N linkage) [43,44]

2.3. Formulation of nanodispersions

Nanolubricants were prepared with 0.05 wt% of nanopowders (rGO or rGO@ODA) and 1 wt% of ionic liquid ($[P_{4,4,4,2}]$ $[C_2C_2PO_4]$) in the base oil (BIOE). Thus, the following nanodispersions were prepared: BIOE + 0.05 wt%, BIOE + rGO, 0.05 wt% rGO@ODA, BIOE + 1 wt% IL + 0.05 wt% rGO and BIOE + 1 wt% IL + 0.05 wt% rGO@ODA.

For the nanodispersions IL-free a typical two-step procedure was used: dry nanopowders were mixed with the BIOE oil using a Sartorius balance (readability of 0.01 mg) to determine the mass concentrations. Then, homogenization of the dispersion was performed with a ultrasounds bath (Fisherbrand) with continuous mode during 4 h (180 W power and 37 kHz frequency). Regarding nanodispersions containing IL, a modified two-step method was used [25], instead of directly mixing nanopowders with base oil, in this case adequate amounts of rGO or rGO@ODA nanopowders were added to an agate mortar and mechanically mixed with the proper quantity of IL for 10 min. Afterwards, this melt was added to the BIOE oil and homogenized as in the previous method.

2.4. Friction tests

Friction rotational tests were carried out using a CSM Standard tribometer operating in ball-on-disc arrangement for the BIOE prepared nanolubricants and base oil at room temperature for: 20 N working load (1.8 GPa maximum contact pressure), 3 mm radius trajectory, distance of 340 m and sliding speed of 0.10 $m{\cdot}s^{-1}$. The specimens used are balls of chrome steel (AISI 52100/535A99; 3 mm of radius; hardness: Rockwell Scale 58–66; Ra = 0.05 μm) and stainless-steel discs (AISI52100/535A99; diameter of 10 mm; roughness of 0.02 μm (Ra) and 190–210 Hv30 hardness).

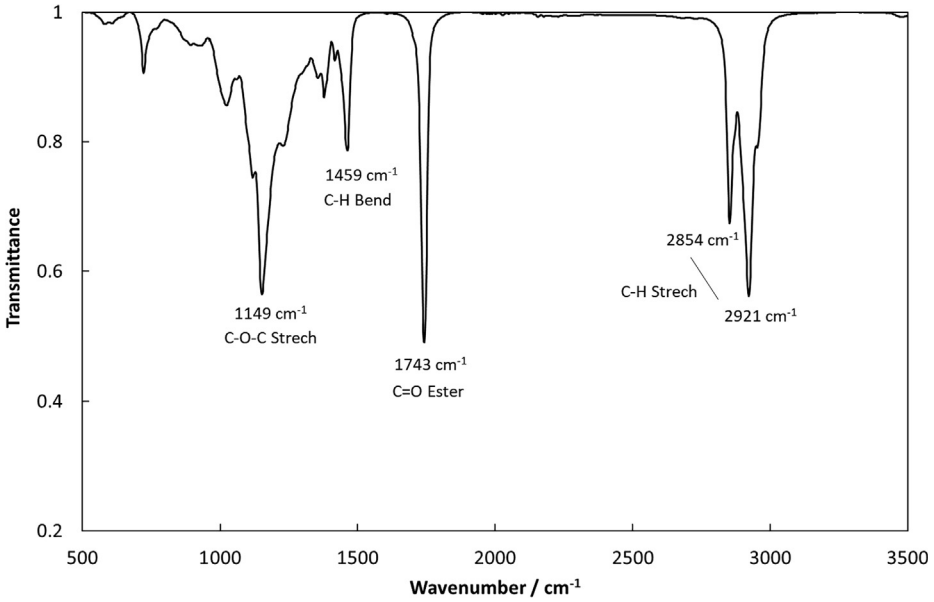


Fig. 1. FTIR spectrum of BIOE base oil.

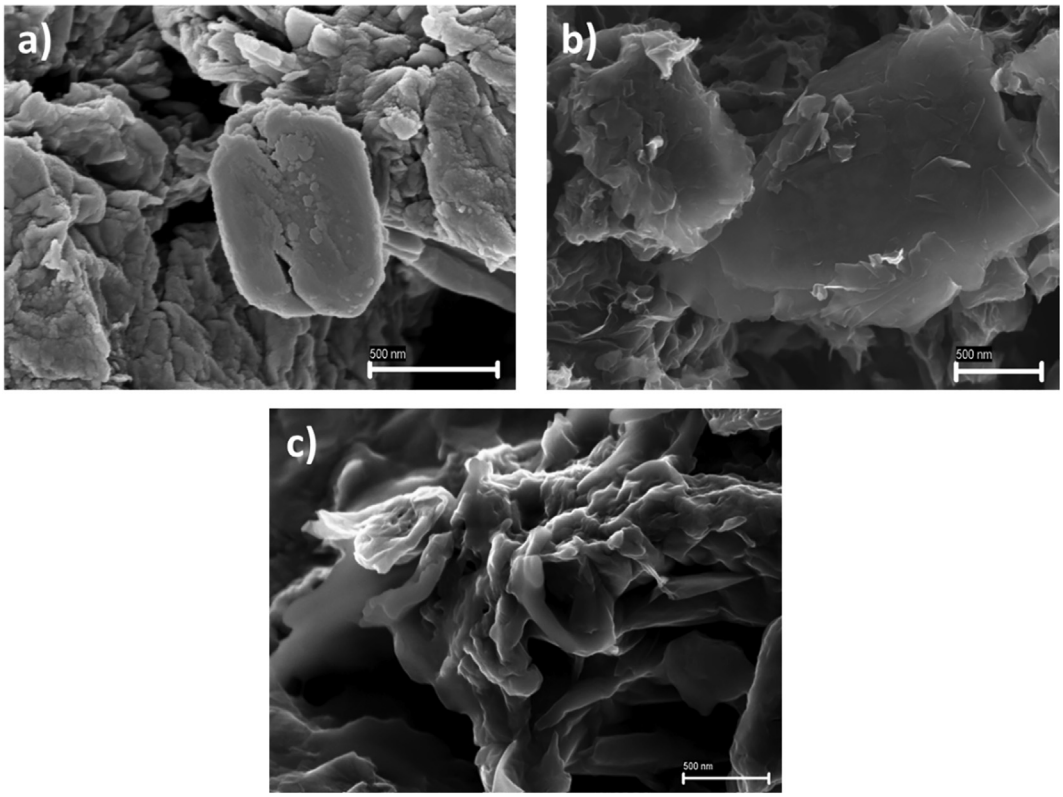


Fig. 2. SEM microimages of GO (a), rGO (b), and rGO@ODA (c) nanopowders [28].

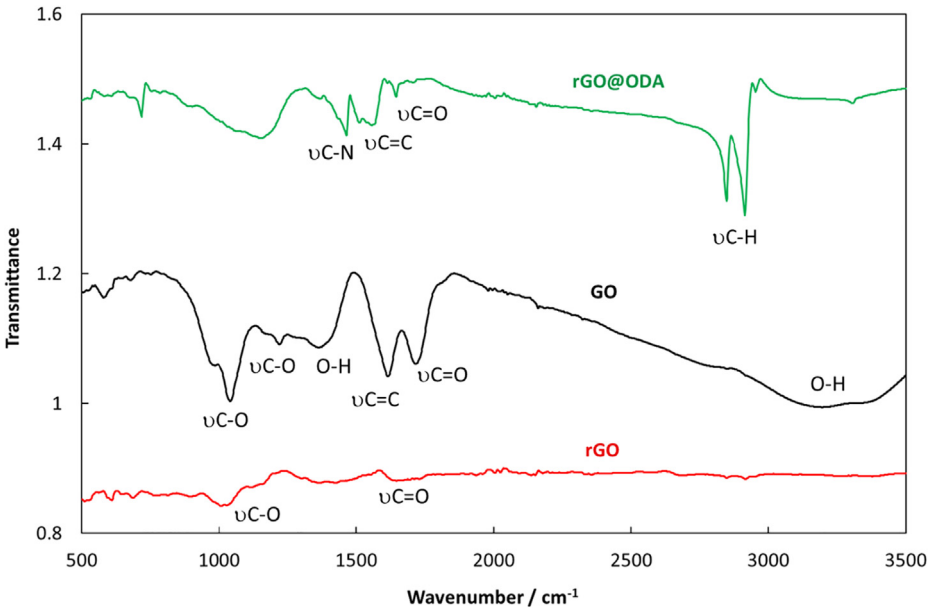


Fig. 3. FTIR spectrum of GO, rGO and rGO@ODA nanopowders showing their characteristic vibrational bands.

Previously to each friction tests, both ball and disc were washed in an ultrasounds bath with acetone to remove any residual material that can affect the friction tests and dried with hot air. Successively, each disc was lubricated with around 0.2 mL of the lubricant under study. It should be noted that at least three replicates were performed to ensure a suitable repeatability. After the friction tests were completed, discs were cleaned with a hexane stream and dried with hot air, being ready to analyze the worn surface. For this aim, a 3D Sensofar S Neox Optical Profiler (confocal method, 10 ×) was used to quantify the wear produced on discs in terms of wear track width (WTW), wear track depth (WTD) and transversal worn area. These three wear parameters were evaluated in three separate regions of each worn track to get representative average values.

The 3D profiler is also used to determine the worn surfaces roughness (Ra) of the discs employed in friction tests to

characterize the anti-wear ability of each nanolubricant. For this purpose, ISO4287 standard was used applying a Gaussian filter (cut-off: 0.08 mm wavelength).

Additionally, a Raman microscope WITec alpha300R+ was used to analyze the surface inside the worn tracks and get information about the distribution of the nanolubricant components (base oil, nanopowders and ionic liquid) on the worn surface, as well as the tribological mechanisms that take place.

3. Results and analysis

3.1. Temporal stability of nanolubricants

The temporal stability of the formulated nanolubricants has been evaluated by means of photographs. This procedure consists of analyzing the deposit of nanoparticles during time. Fig. 4 illustrates that for all nanolubricants there is no sedimentation before three weeks after the nanolubricant formulation. The observed stability is much longer than that needed to complete the friction tests (over 3 h per each lubricant). It can also be inferred from Fig. 4 that for IL-free nanolubricants, sedimentation begins to occur at four weeks. However, for hybrid nanolubricants this fact does not occur, therefore IL clearly improves the stability of the nanopowders (rGO and rGO@ODA) in the oil.

Likewise, refractometry was also used to study the temporal stability of the nanolubricants. The refractive index (n) of nanolubricants was measured every hour up to 50 h, examining its progress (Fig. 5). In a previous research [28] the n evolution of PAO 40 (or of an ester) nanolubricants additived with GO or with rGO was examined, finding that n rose over 0.4% and 0.1% for GO and rGO nanolubricants, respectively, which implies that the modification of nanoparticles improves the stability of nanolubricants. Furthermore, in other previous research [24], the progression of the n

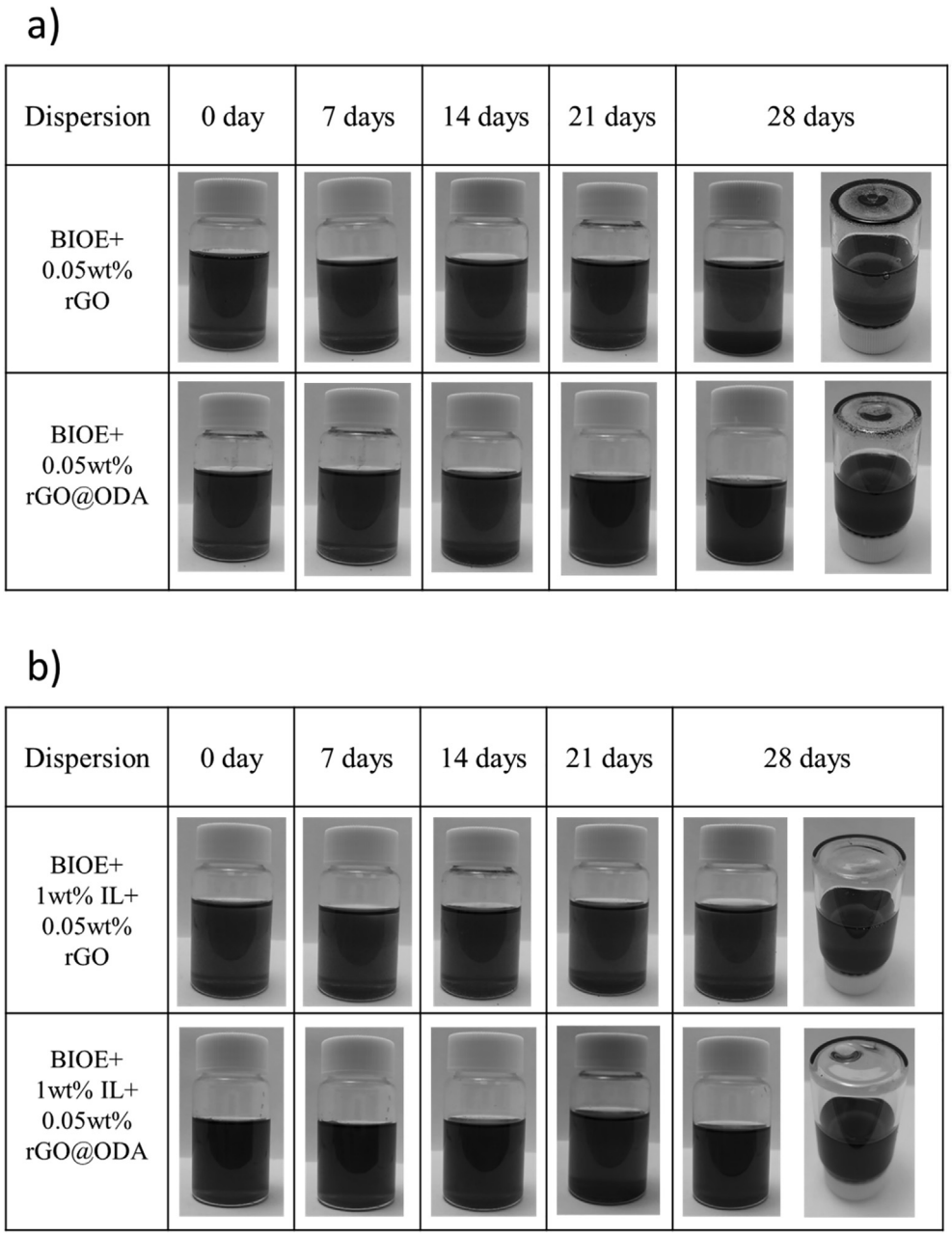


Fig. 4. Stability visual observation of BIOE nanolubricants with (b) and without (a) IL.

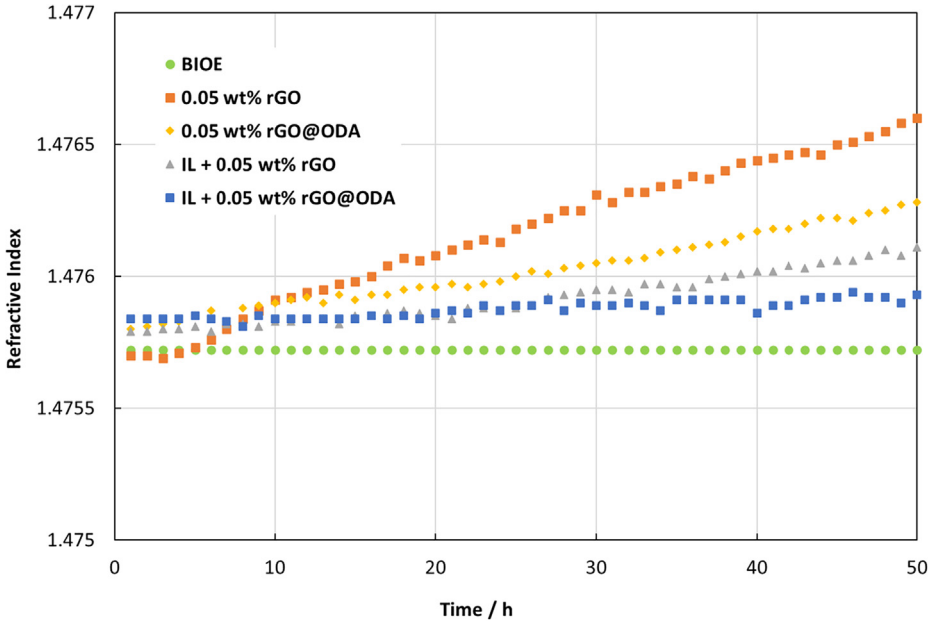


Fig. 5. BIOE lubricants refractive index at 20 °C versus time.

values proves an increase of over 0.1 and 0.2% for (h-BN or GnP) nanolubricants containing and non-containing IL, respectively, showing that the IL addition also improves the stability of the nanolubricants.

In the present study, the refractive index inspection of bio-nanolubricants (Fig. 5) reveals increases around 0.01 and 0.02% for 1 wt% IL + 0.05 wt% rGO@ODA and for 1 wt% IL + 0.05 wt% rGO nanolubricants, respectively. Regarding the nanolubricants without IL, growths in refractive index of 0.03 and 0.06% were obtained for 0.05 wt% rGO@ODA and for 0.05 wt% rGO nanolubricants, respectively. The achieved results reveal a great stability for the designed nanolubricants, especially in the case of nanolubricants that contain IL. Therefore, it can be concluded that the combination of chemical modified nanopowders and the addition of IL as dispersant is an excellent alternative to achieve a better stability of nanolubricants.

3.2. Tribological characterization

In Fig. 6 and Table 1 it is showed the average coefficients of friction (μ) for the formulated BIOE nanolubricants and for the BIOE base oil. It is visibly observed that the coefficient of friction decreases considerably for all nanolubricants compared to that of BIOE base oil. Specifically, the best antifriction performance occurs for the hybrid nanolubricant BIOE + 1 wt% IL + 0.05 wt% rGO, showing a friction coefficient of 0.0682 versus 0.1027 obtained for the non-additivated BIOE (Table 1), which leads to a 34% friction reduction. Regarding the other designed nanolubricants, friction reductions of 26% (BIOE + 1 wt% IL + 0.05 wt% rGO@ODA), 21% (BIOE + 0.05 wt% rGO) and 14% (BIOE + 0.05 wt% rGO@ODA) were achieved.

Concerning the wear produced during friction tests, 3D mapping and cross-sectional profiles of worn scars in lubricated discs

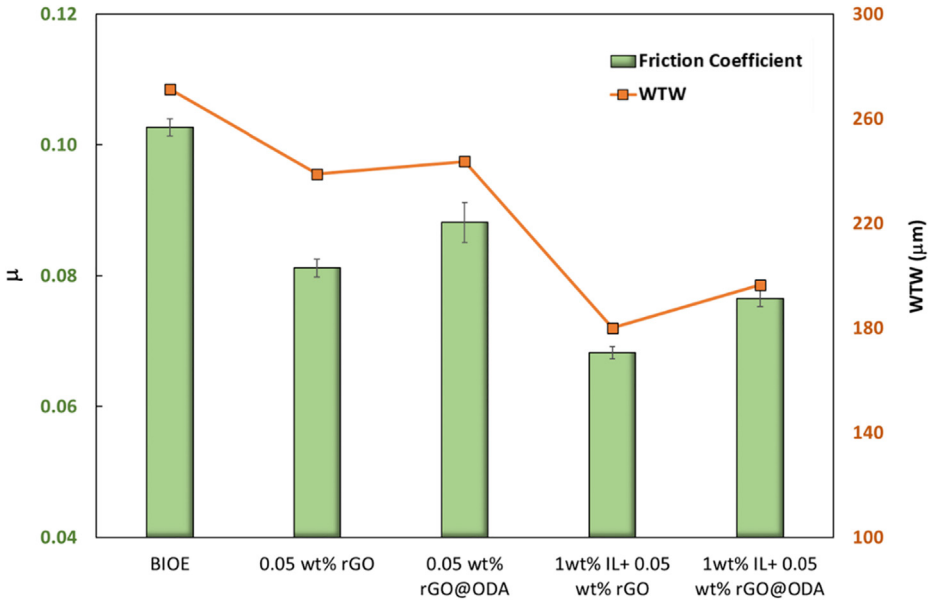


Fig. 6. Comparison between the friction coefficients (μ) and the wear track widths (WTW) achieved with BIOE and BIOE nanolubricants.

Table 1
Average coefficients of friction, μ, and average wear parameters: width (WTW), depth (WTD), and area as well as their standard deviations for the tested lubricants.

Lubricant	μ	σ	WTW/μm	σ/μm	WTD/μm	σ/μm	Area/ 10^2μm^2	σ/ 10^2μm^2
BIOE	0.1027	0.0014	271	14	1.38	0.14	2.06	0.25
+ 0.05 wt% rGO	0.0811	0.0013	239	13	0.93	0.12	1.10	0.16
+ 0.05 wt% rGO@ODA	0.0881	0.0031	244	11	1.10	0.12	1.46	0.13
+ 1 wt% IL + 0.05 wt% rGO	0.0682	0.0010	180	10	0.61	0.11	0.59	0.17
+ 1 wt% IL + 0.05 wt% rGO@ODA	0.0765	0.0012	196	12	0.62	0.10	0.70	0.13

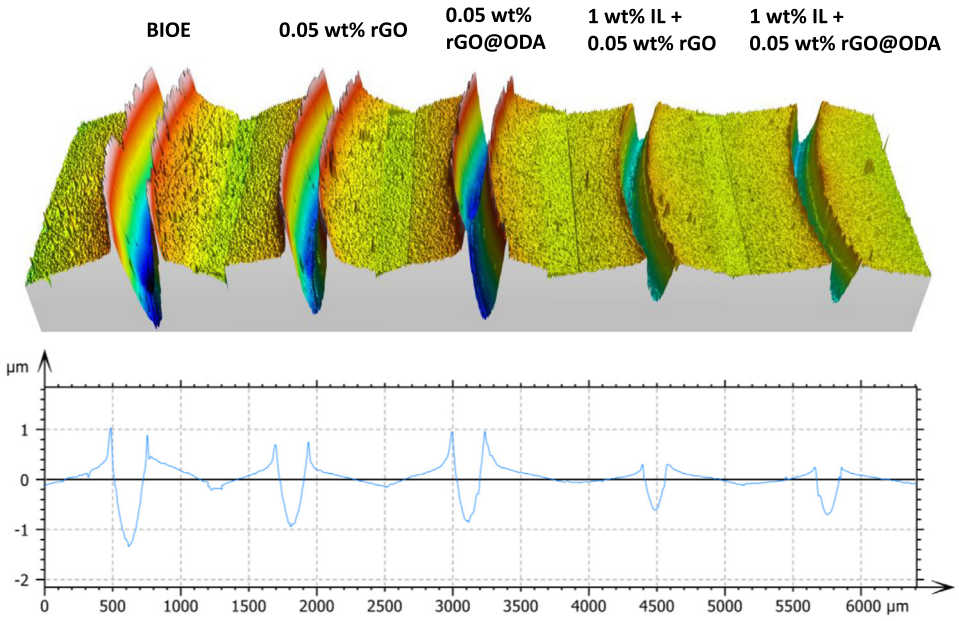


Fig. 7. 3D Surface topography and worn scars cross section profiles for all the BIOE lubricants.

are displayed in Fig. 7 and Fig. 8. The WTW, WTD and cross-sectional area values found for the worn tracks on the discs lubricated with each lubricant are presented in Table 1. It can be highlighted that for all the nanolubricants, the produced wear is less than for the BIOE base oil for all the wear parameters (WTW, WTD and area). The maximum wear decreases in width and area were achieved with the BIOE + 1 wt% IL + 0.05 wt% rGO nanolubricant, with 34 and 71% reductions, respectively. As regards the

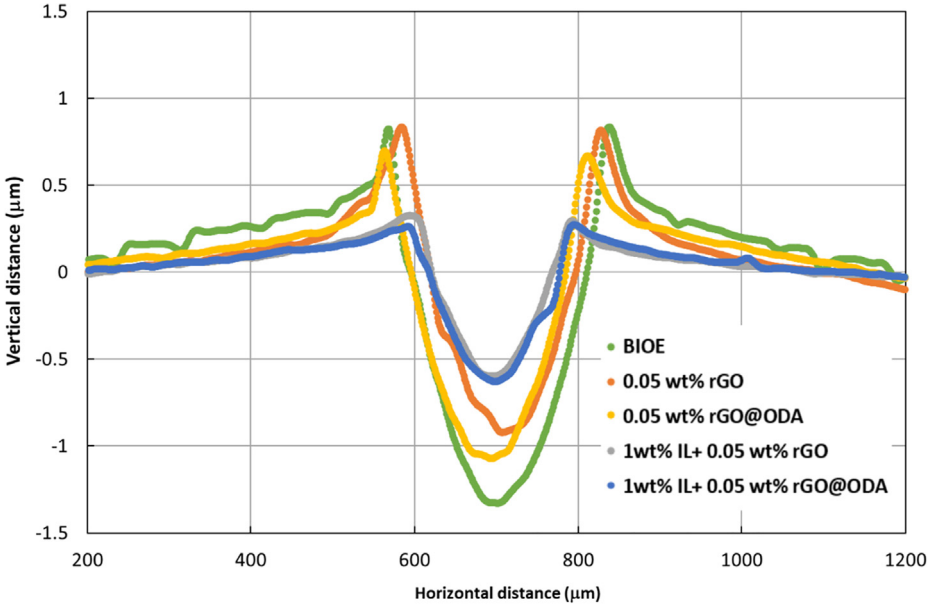


Fig. 8. Cross section profiles comparison of worn scars for the BIOE lubricants.

Table 2
Values of roughness, Ra, and their uncertainties, σ, in the worn scars lubricated with tested BIOE nanolubricants and BIOE using a Gaussian filter (0.08 mm cut-off).

Lubricant	Ra/nm	σ
BIOE base oil	210	22
+ 0.05 wt% rGO	150	19
+ 0.05 wt% rGO@ODA	148	14
+ 1 wt% IL + 0.05 wt% rGO	112	10
+ 1 wt% IL + 0.05 wt% rGO@ODA	124	13

WTD, the lowest values were obtained with the BIOE + 1 wt% IL + 0.05 wt% rGO and BIOE + 1 wt% IL + 0.05 wt% rGO@ODA nanolubricant with reductions of 56% and 55% compared to BIOE oil.

Roughness (Ra) of the worn disc's surfaces were examined to investigate the antiwear nanolubricants capability. Table 2 shows that worn tracks lubricated with each of the studied nanolubricants have smaller roughness than those lubricated with BIOE base oil. In particular, a Ra value of 210 nm was achieved in the BIOE worn scar while for the track lubricated with the BIOE + 1 wt%

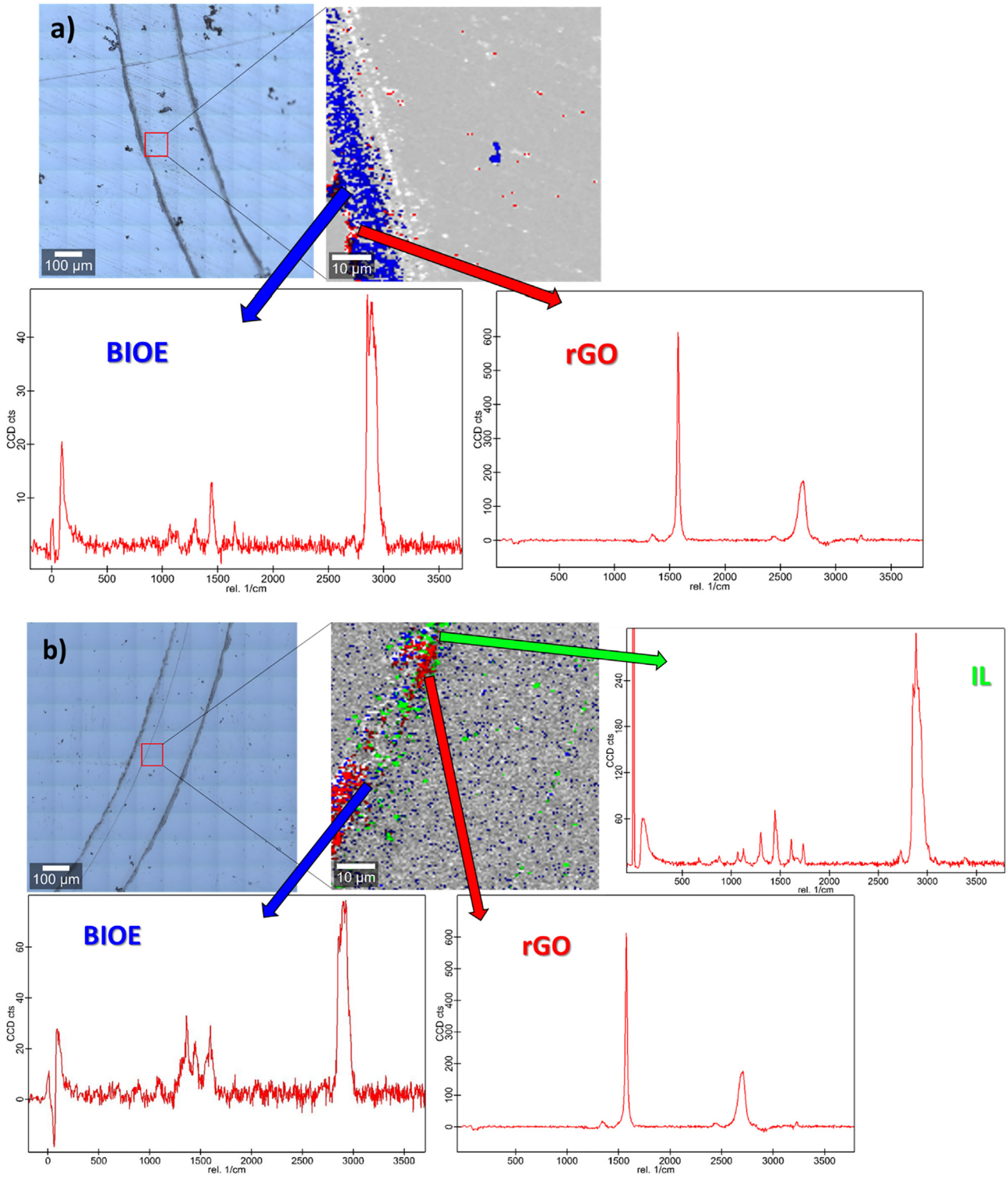


Fig. 9. Elemental map and Raman spectra of worn surfaces tested with rGO nanolubricants a) BIOE + 0.05 wt% rGO and b) BIOE + 1 wt% IL + 0.05 wt% rGO.

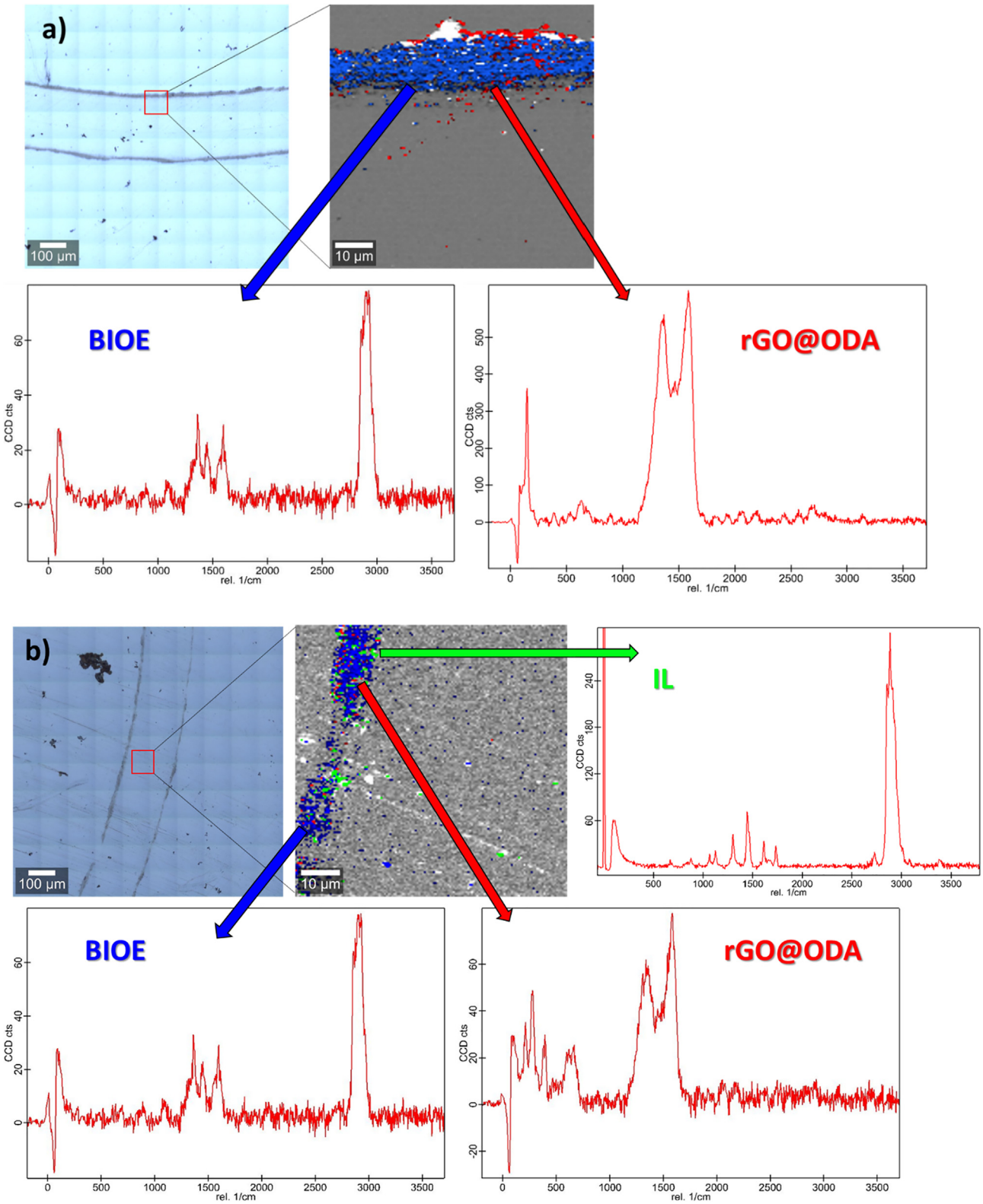


Fig. 10. Elemental map and Raman spectra of worn surfaces tested with rGO@ODA nanolubricants a) BIOE + 0.05 wt% rGO@ODA and b) BIOE + 1 wt% IL + 0.05 wt% rGO@ODA.

IL + 0.05 wt% rGO nanolubricant the lowest Ra value (112 nm) was reached, which implies a roughness reduction of 53%.

Raman spectra and elemental mapping of the worn scars lubricated with BIOE nanolubricants additived with rGO or rGO@ODA with or without IL were taken through a confocal Raman microscope (wavelength of 532 nm) to know the role that additives play in the wear reduction. Raman spectra of nanolubricants components (BIOE, IL, rGO and rGO@ODA) are exhibited in

Figures S1-S4, respectively, observing that their typical peaks agree with some of those located on the worn discs lubricated with nanolubricants. Fig. 9a and Fig. 10a show a substantial presence of BIOE (blue color) and the existence of spots due to rGO and rGO@ODA nanopowders, (red color) on the worn track, especially at the edges. Fig. 9b and Fig. 10b, corresponding to the discs lubricated with BIOE + IL + rGO and BIOE + IL + rGO@ODA, illustrate the presence of BIOE (blue color), rGO or rGO@ODA nanoadditives (red color) and IL (green color) in the worn track, the additives position in worn tracks being parallel to friction sliding lines. This fact shows that nanopowders can improve the lubrication capacity of the non-additivated BIOE by producing thin tribofilms on the tribo-stressed area. In addition, van der Waals interactions between the rGO or ODA-rGO nanosheets offer low shear strength under sliding contact stress, causing a reduction in friction, whereas the constant stock of these nanomaterials in the tribological contact avoids the contact between sliding surfaces and decreases wear. Berman et al. [45] showed that 2D graphene derivatives form a protective tribofilm on the contact interfaces that provides easy shearing, resulting in friction and wear reductions. Taking this into account and that the worn surface after tests with hybrid and non-hybrid nanolubricants of rGO or rGO@ODA are smoother than that with BIOE base oil (Table 2), as well as the corresponding Raman analyses, it can be supposed that tribological mechanisms are the tribofilms formation as well as the repairing surface and the synergistic effects due to nanoparticles and IL.

4. Conclusions

In the present research the following conclusions were reached:

- Two reduced graphene oxide nanopowders, reduced graphene oxide (rGO) and octadecylamine-modified reduced graphene-oxide (rGO@ODA), have been synthesized.
- Four nanodispersions based on a biodegradable ester BIOE were prepared, two with reduced graphene oxide (rGO or rGO@ODA) nanomaterials and two hybrid nanodispersions using an IL and rGO or rGO@ODA as additives, showing good stability times of at least three weeks.
- Friction coefficients achieved for all the nanolubricants are lower than that found for the BIOE base oil, being the best friction reduction (34%) for the 1 wt% IL + 0.05 wt% rGO nanolubricant.
- For all the nanolubricants (hybrid and non-hybrid) the generated wear in the discs is lower than that for the neat BIOE in width and area. Specifically, the highest reductions were achieved for the BIOE + 1 wt% IL + 0.05 wt% rGO nanolubricant with 34% in WTW and 71% in area.
- Through roughness and Raman studies, surface repairing, synergistic effect and formation of tribofilm mechanisms owing to the nanopowders and IL were validated.

CRediT authorship contribution statement

José M. Liñeira del Río: Writing - review & editing, Writing - original draft, Methodology, Investigation, Conceptualization. **Enriqueta R. López:** Writing - review & editing, Validation, Supervision, Formal analysis. **Fátima García:** Writing - review & editing, Methodology, Investigation. **Josefa Fernández:** Writing - review & editing, Validation, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.molliq.2021.116885.

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